FirstPass STUDIO

USER GUIDE

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

VERSION 1 · OCTOBER 2026 · BY PIVT

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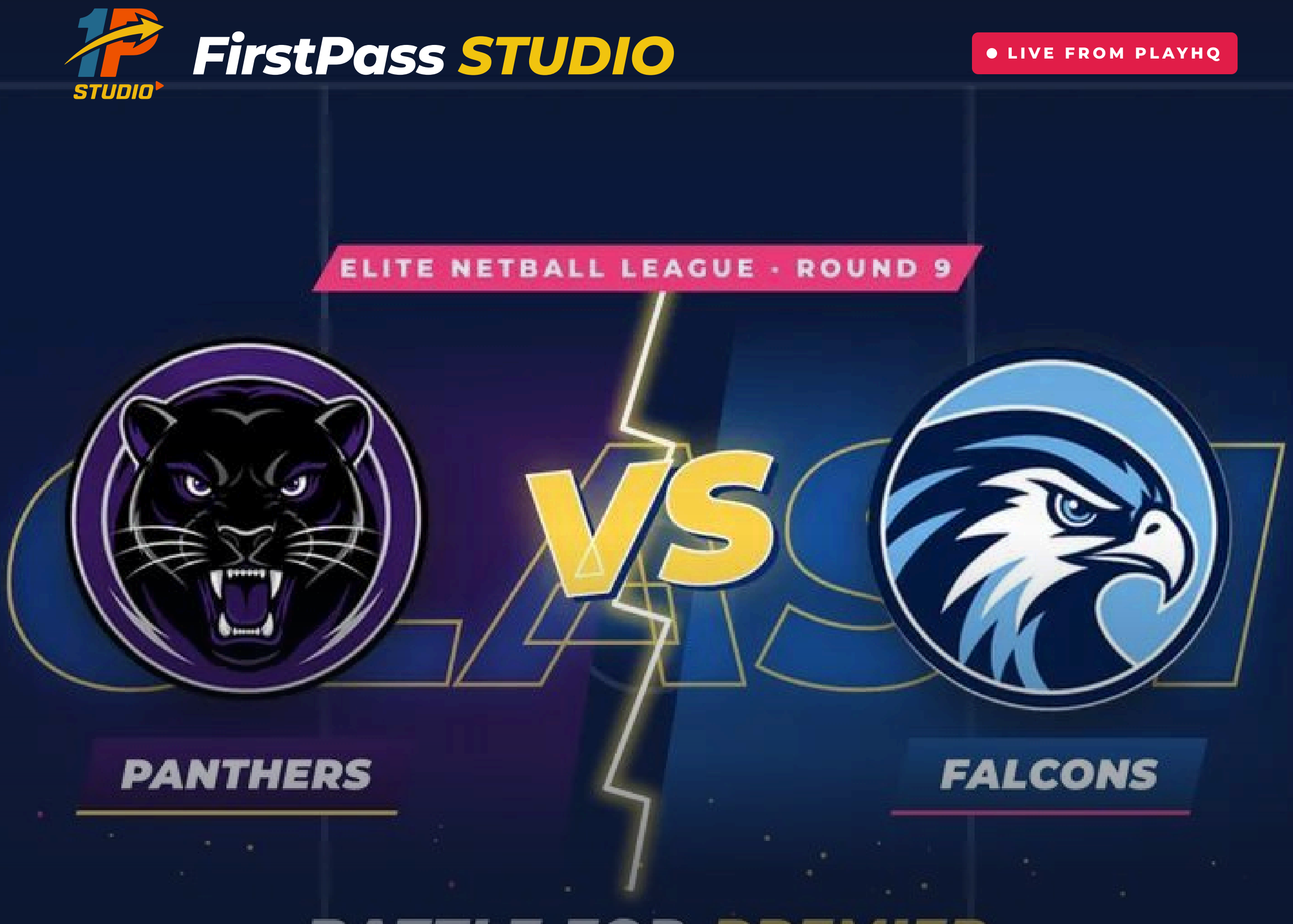

USER GUIDE

Graphics that follow the game.

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

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ABOUT THIS GUIDE

Graphics that follow the game.

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

5 s
The live score, quarter and clock read from PlayHQ every 5 seconds while the game is scored.

31
Broadcast graphics, from the scoreboard and ladder to the sponsor wall, plus a test pattern.

17
Competition logo animations, and six intro and outro styles.

1
Operator on a followed night. Sequences run the run-up, the breaks and full time.

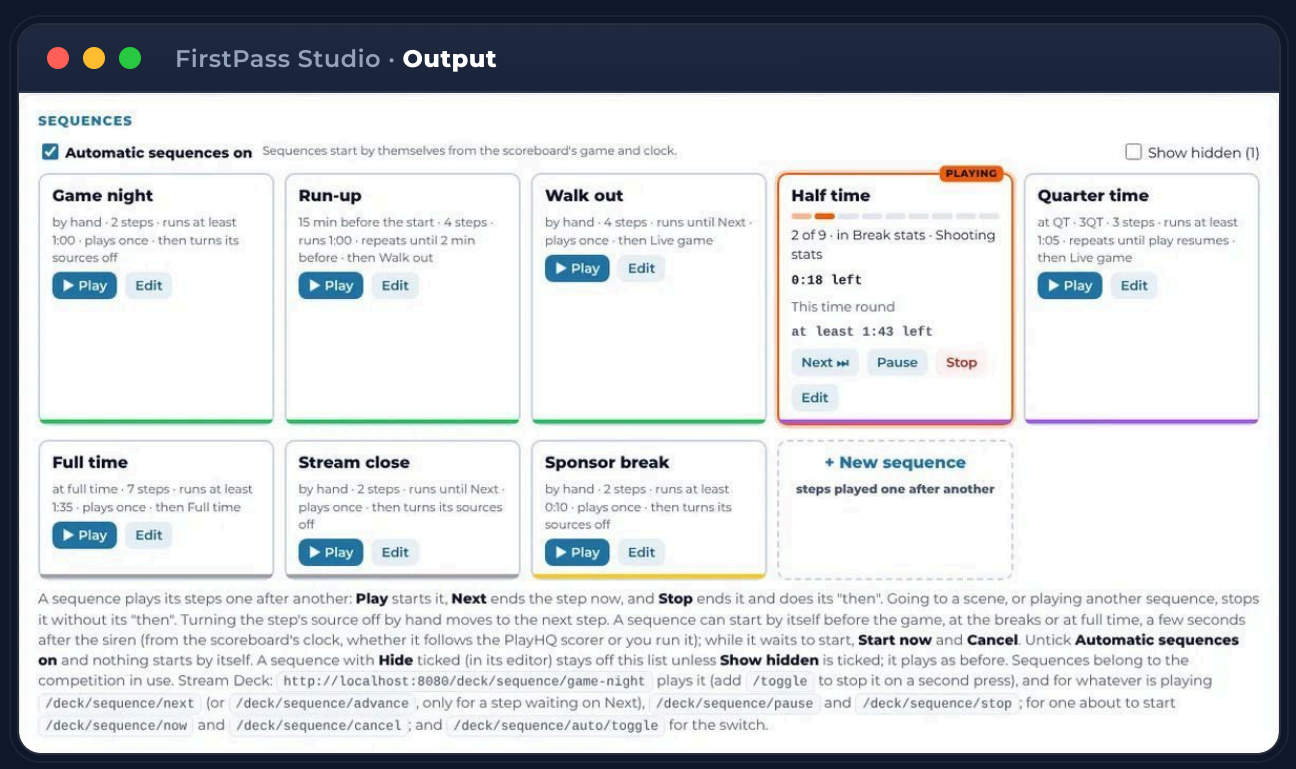
IN
PlayHQ
The fixture, ladder, line-ups and the live score from the scorer's tablet. No account, no key.

→

ON YOUR MAC
FirstPass Studio
Builds one 1920 × 1080 scene from layers: the base scene, automatic layers, scenes and sequences.

→

OUT
Your switcher
A display, a Blackmagic device or NDI, with fill and key or a chroma colour over the cameras.



The Output page runs the show; the super shot plays by itself.

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Every screenshot comes from Studio running the fictional Elite Netball League: eight clubs named after animals, three divisions, a live game and fictional sponsors.

CHAPTER 1 OF 11

What Studio is

01

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

1. The words used in this guide
2. The pages you work in
3. The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

1.1 The words used in this guide

Studio
The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.

On air, off air
Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.

A graphic, or a page
One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.

A source
A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.

The scene
Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.

A scene (on the Scenes list)
A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.

A sequence
A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.

The competition
Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.

The data folder
Where Studio saves everything, outside the app, so updates never touch it.

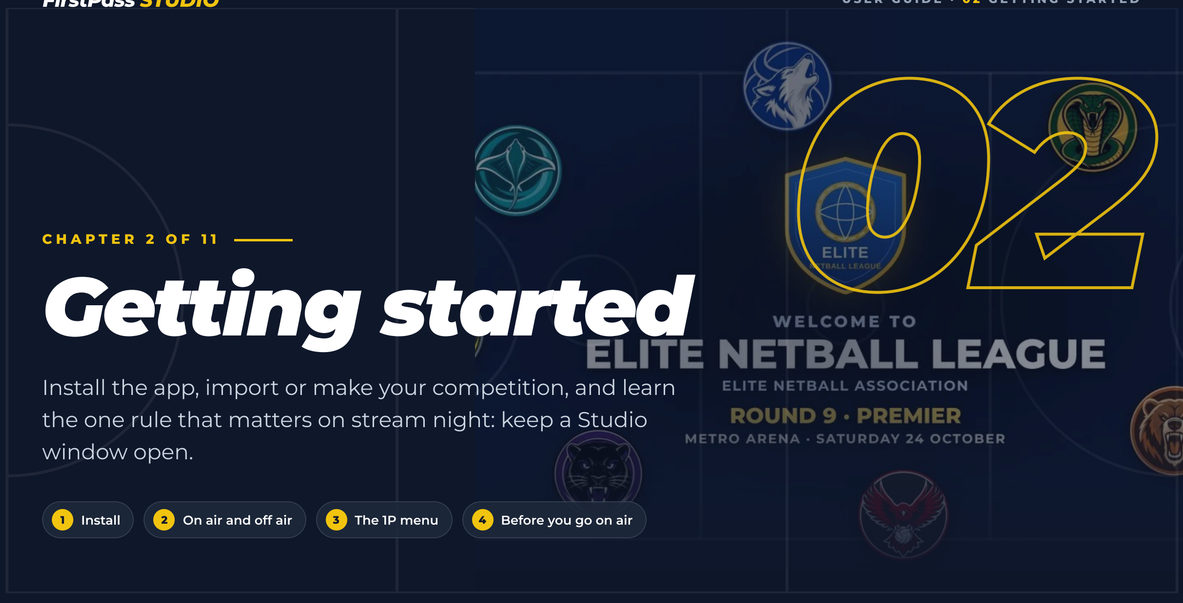
1.2 The pages you work in

Every Studio window has the same tab bar: **Output**, **Scoreboard control**, **Competitions setup**, **Sponsors**, **Athletes**, **Stream Deck** and **Docs** (⌘1 to ⌘7 in the app). Output is where the show is built and sent out; Scoreboard control runs the board; the rest are set up before the night.

> ***Studio does one job on stream night: graphics.***

1.3 The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.



CHAPTER 2 OF 11

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

1 Install · 2 On air and off air · 3 The 1P menu · 4 Before you go on air

2.1 Install

01
Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).

02
The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.

03
Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).

04
Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

2.2 On air and off air

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The status line under the Output page's title shows the game on the board, whether scores and clock follow PlayHQ, how long since PlayHQ last answered (amber when overdue, red after a minute of errors) and whether the video output is on. If anything is wrong, this line says so on the page you keep open.

2.3 The 1P menu

Open Studio, Scoreboard control, Competition setup, Copy the Studio link, Install the Stream Deck plugin, Back up… and Restore…, Settings folder (iCloud Drive), Open the settings folder, Open the debug log, Restart Studio, Open at login, Let other devices change things, What's new…, Check for updates, Stop Studio (while on air) and Quit. Updates download in the background and install when the app quits, never mid-stream; **Restart to update** installs one straight away.

LINKS Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/`, so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

> **The one rule that matters on stream night: keep a Studio window open.**

CHAPTER 2 · GETTING STARTED

2.4 Before you go on air

01
Open the app. The Output page opens and Studio goes on air. Keep a window open all night.

02
Right competition. Competitions setup shows the one in use at the top. Check its PlayHQ link, the court to follow and the scoring window.

03
Right game. On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.

04
Scores arrive. Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.

05
Scene and output. On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.

06
Sound. Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.

07
Stream Deck. Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.

08
Sponsors. Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

CHAPTER 3 OF 12 · Introduction

Game night, the automated way

This is how most nights run: one operator, PlayHQ sounding out game, a base scene that stays on, automatic layers that fire by themselves, and sequences that handle the run-up, the breaks and full time.

How it fits together

Run through do this work, and each one is set up once.

Follow PlayHQ. The operator follows the competition, scoreboard, and clock from the same feed the PlayHQ game page uses, every 5 seconds by default. Nothing is typed in by hand. Everything else in the system reacts to what PlayHQ says is happening, so getting PlayHQ right is the single most important thing you do.

The base scene. One scene is left on for the whole run of play, run through the game: the scoreboard, the live banner, the branding logos and the ticker. It never changes.

Automatic layers. Sponsor boards that slip on and off by themselves, a countdog happens the super-short celebration, the hot streaks and goal flashes. Once you've set them up, you don't touch them again.

Sequences. Steps played in order for the parts of the night that aren't just "live": the pre-game run-up, quarter and half breaks, and full time and the close. A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.

> A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.
>
> CHAPTER 3 · GAME NIGHT, THE AUTOMATED WAY

Scoreboard control

Add a game sets the two teams and starts the clock. **Swap home and away** flips the sides if the teams change ends. **Reset the scoreboard** clears the game and zeroes everything. **Clock** starts, pauses, resets, and nudges the clock by a minute or a second. **Period** moves between periods and sets the period number. **Scores** adds or takes away a point for either team, or sets a score directly. **Penalties** records a penalty, edits one, or removes it. **Timeouts** uses a timeout, cancels one, or sets how many each team has left. **Undo** and **Redo** step back and forward through changes, and **Undo all changes** and **Redo all changes** jump straight to the start or the end. **Match** brings a game in from PlayHQ or sends the result back, or clears the match.

PRO TIP Scoreboard control lets you set the clock, the period, and the scores by hand, so you can run a game even if PlayHQ is down. Undo and Redo mean a mistake is one click to fix.

Clocks doesn't see PlayHQ but feed for a game with its electronic scoring window open, whether the game was played automatically or by hand. Clocks can't tell and will start checking.

The Output page

Add output in the Studio graphics list or any rate switcher. **Add Image** (PNG, JPG, GIF, WebP or SVG) and **Add video** (MP4, WebM or MOV). All media goes into a shared library, so each file is uploaded once and can be used on any output. Drag a source up or down the list to move it in front of or behind others. **Sound output** picks which one's audio is heard.

Source settings is where you point a source at what it shows. For a lower third, the board shows the address of the page to load, the title text, and the competition logo and position. For a sponsor board, it shows the sponsor and the page size. For a video, it shows the file, how long to fade in and out, and where it sits on the screen. **When shown** controls whether a source appears all the time, only when live, or only when not live.

Image and **Video** each show the files you've added, with a thumbnail and the file name. Click a file to select it, or drag it up or down to change the order it's shown in. **Add image** or **Add video** opens a file picker, and you can also drop files straight onto the panel. **Remove** takes the selected file out of the library.

Lower third is the broadcast graphic for a live game. Its **Source settings** point it at the scoreboard and the home sponsor and position, logo, and scale it, and set when it shows.

Sponsor boards is the rotating strip for the home sponsor. Add a sponsor and drop in their image or video, and it joins the rotation. Remove one when the sponsor's season is over.

Countdog, **Hot streak**, and **Goal flash** each take a video for their moment. Countdog's video plays over the last few seconds of a quarter or game, Hot streak's for a team on a run, and Goal flash's for a goal. Drop a file in and it's ready to fire.

Quarter time, **Half time**, and **Full time** each take the video for their break. Quarter time's video plays over the end of a quarter, Half time's over the break, and Full time's over the close of play.

Sound output picks which one's audio is heard. Only one source plays sound at a time, so choosing one here mutes the rest. Sources without audio are skipped over here.

The base scene

The base scene is left on for the whole run of play.

In the sample league it is the **Live game** scene the operator switches to when the game starts. It holds the scoreboard, the live banner, the branding logos, and the ticker, and it never changes. Switch to it at the start and it stays on for the whole night.

Scoreboard. Count in a pink box on the left, both teams and their scores, and the game clock. Set it to show the period, or to hide it. The scoreboard's fair shows while a game is live, and PlayHQ's "Powered by" tab carries the naming right sponsor.

Title. Sounds, tournament and venue, with the naming right sponsor's logo alongside.

Branding logos. The club's and league's logos, in the top-right corner.

Score ticker. Scrolling scores from the other courts, fed by a Partner and up-sponsor's logotic. Its speed is set on the Output page, in pixels per second.

Build it once and leave it on. Everything else in **what's on now** under Scenes and name it. From then on, one click on Live Stream Deck key brings the show back to this look.

Automatic layers

Layers are the things that come and go by themselves during the scoreboard. They all add to PlayHQ sequence of their own: they set Studio, which already has the scene.

Super shot. Super shot's video plays over the scoreboard. Set it to start with the game, and it will play once and then stay off. Turn the layer off to hide it.

Hot streak. Showing up on a game's a run, for her on a Goal for. Set it to start with the game, and it will play once and then stay off. Turn the layer off to hide it.

Goal flash. Goal flash's video plays over the scoreboard. Set it to start with the game, and it will play once and then stay off. Turn the layer off to hide it.

Each celebration can carry a sponsor's "Brought to you by" tag (Chapter 7). A vibration-short super shots plays to super shot animation and counts to super shots a set on the competition set-up page, per division if needed.

Scenes

A scene is a saved set of sources on and off. Clicking a scene goes to it, turning sources on and off to match while anything already on that's also in the new scene stays where it is. Scenes are listed for the order in the order of green arrows, and you can drag them to reorder.

Live game is the one the operator switches to when the game starts. Typical layout is source on it is a your shows it's on, and the scoreboard, lower third, sponsor boards, and ticker are always on.

Pre-game is for the run-up before a game. Your usual layout is source on it is your shows it's on, and the scoreboard, lower third, sponsor boards, and ticker are always on.

Quarter break is for the breaks between quarters. Your usual layout is source on it is your shows it's on, and the scoreboard, lower third, sponsor boards, and ticker are always on.

Half time does the same until on last and the water lower third and sponsor boards, and goal boards for in-game scores since it. Your usual layout is source on it is your shows it's on, and the scoreboard, lower third, sponsor boards, and ticker are always on.

Full time does the same until on last and the water lower third and sponsor boards, and goal boards for in-game scores since it. Your usual layout is source on it is your shows it's on, and the scoreboard, lower third, sponsor boards, and ticker are always on.

Next game is for between games. Your usual layout is source on it is your shows it's on, and the scoreboard, lower third, sponsor boards, and ticker are always on.

Clear hides everything. Use it when the ruins with the night's teams. Turn the output off. The operator report has the night's scorecard ready to send.

Sequences

Sequences run the game's run-up, breaks and full time. Each step turns on the sources you pick and plays the s set time, until it finishes (the video ends or its animation entrance ends), until the game moves on (the clock starts, or the next break or full time), or by hand. Any step can be skipped, and the whole sequence the next step doesn't open up off with their own timer.

Pre-game, **Quarter break**, **Half time**, and **Full time** are the four sequences you'll use. Each one starts with no steps, and you add them to it in the order you want them to play.

Add step adds a step to the sequence. Pick what it does from the list: show or hide a source, play a video, or wait for a set time. Each step has a **Duration** box for how long it runs, and **Remove step** takes it out.

Show and **Hide** turn a source on or off. **Play video** plays a video source. **Wait** pauses for a set time, **Wait for clock** for the game clock to start, **Wait for video** for a video to finish, and **Wait for animation** for an animation to finish. **Duration** sets how long a step runs.

Duration sets how long a step runs, in seconds. Leave it blank on a step that should run until the next step moves on.

Remove step takes the selected step out of the sequence. Add them back if you change your mind.

Music and playlists

Music
Firethorn Studio
Music
Add playlist
Drop audio files here, or click to choose files
Add playlist
Name
Pre-game warm-up
Tracks
Warm-up mix
Track 1.mp3
Track 2.mp3
Add a song
Cancel
Add playlist
Add song
Add playlist
Add song
Add playlist
Add song

CHAPTER 4 OF 11

Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

1 Two people, not one
2 Switching to manual
3 The score
4 The clock and the quarter
5 Breaks and celebrations by hand
6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

4.1 Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

Production manager

- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

> **If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks.**
>
> CHAPTER 4 · RUNNING IT BY HAND

4.2 Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again.

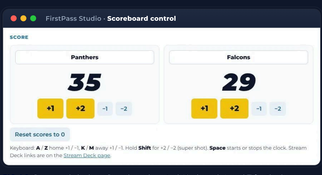


FIG 4.1 Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT

If PlayHQ stops answering during a followed game, the scoreboard keeps its last score and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

4.3 The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

4.4 The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.

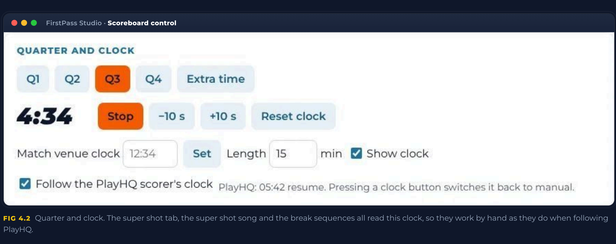


FIG 4.2 Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

4.5 Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.
- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

4.6 Finals

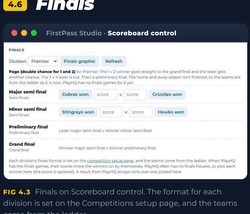


FIG 4.3 Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.

CHAPTER 5 OF 11

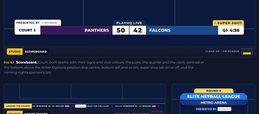
The graphics

Every graphic Studio draws, grouped by when it is used. Each one is added to the scene with **Add page** on the Output page, and its options appear as fields under its add-on. Options named here are the ones worth knowing.

- always on
- Before the game
- During play
- After the game
- No competition
- The breaks
- Checking the outputs

All of them follow the game, round and court picked on Scoreboard control, take the competition's colours, font and logos from its setup, and fill the scene's Name, Cards at the top between the title and the scoreboard and stroke to fit when they have more in them than fifty.

5.1 Always on

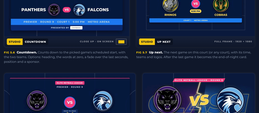


Scoreboard — CLOSE UP · ON SCREEN

5.1.1 Scoreboard. Court, both teams with their logos and club colours, the score, the quarter and the clock, centred at the top of the scene. Options position (top centre, bottom left and so on), lower-third set on or off, and the ticking-lights sponsor's tab.

Opening title — CLOSE UP · ON SCREEN

5.1.2 Scene Order. Title scenes from this country's other games existing among the scenes, not just this one's. The game-only scenes, updated every 60 seconds. Speed from the control page or a Smash Cans key; a Preview sponsor's logo takes a slot in the line.



Branding logos — CLOSE UP · ON SCREEN

5.1.3 Branding logos. PIVT and FirstPass Studio left of the title, the competition logo right. Two-second fades, plus Hight and Hold about the placement.

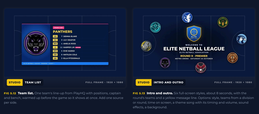
Viewers — FULL FRAME · TWO / TWO

5.1.4 Viewers. Its watching! From the competition's YouTube channel, or the Studio's own count. Options **Minimum viewers** to hide it below a threshold, and **Show zero** to keep it a port for the end of the night.

5.2 Before the game

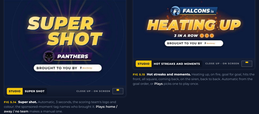
Countdown — CLOSE UP · ON SCREEN

5.2.1 Countdown. Counts down to the game's scheduled start, with the two teams, the commentators heading, the words at zero, a fade over the last seconds. Options and a sponsor.



Match up — CLOSE UP · ON SCREEN

5.2.2 Match up. Both teams side by side with their full line-ups, across the centre. Options and the season's head to head. Fade with **Head to head** off to cash the holders.



Competition match ups — CLOSE UP · ON SCREEN

5.2.3 Competition match ups. Who meets whom, position by position on the court. Options whether the scene is a list or a grid, and if a photograph. Options records are the head-to-head stats, each with its W, the and pairing only, whether away at the start.

Team list — FULL FRAME · TWO / TWO

5.2.4 Team list. The starters line up from Play/HQ with positions, caption and jersey, and it's switched up before the game so it shows all once. Add one source per side.

Toss and duties — FULL FRAME · TWO / TWO

5.2.5 Toss and duties. A 10-second scene, about 8 seconds, with the toss winner and what they've picked, the starting positions, duty reports from C Graders, and a round, time on screen, a theme song with its timing and volume, sound effects, a background.

5.3 During play

Super shot — CLOSE UP · ON SCREEN

5.3.1 Super shot. Automatic, 10 seconds, when the scoreboard's Super Shot is pressed. Options the big name show to go to **Play home / away** if no team makes a manual one.

On the siren — CLOSE UP · ON SCREEN

5.3.2 On the siren. A goal in the last seconds of a quarter. Moments are flagged, and a scene with the scoreboard holding them put, they also take the sponsor's Sponsor.

Last 10 goals — CLOSE UP · ON SCREEN

5.3.3 Last 10 goals. The game's recent goals rolled in newest with a scoreboard, the scoring team highlighted, and the shooter's name and number. Options **Count** to keep it to 10 before the top in down the side, how many given, position and heading.

Person lower third, athlete — CLOSE UP · ON SCREEN

5.3.4 Person lower third, athlete. Name, team and shot tags from the athlete register, with their photo if there is one. The Output page supplies names as you type.

Person lower third, coach — CLOSE UP · ON SCREEN

5.3.5 Person lower third, coach. Name, title and the competition's logo for a commentator, or an official or a sponsor's guest. Options a set above, position set or right, animate away.

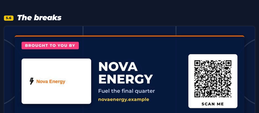
5.4 After the game

Winner — FULL FRAME · TWO / TWO

5.4.1 Winner. The winning team, logo, final score and a stop word, worked out from the board each time it comes on. Options **Stop** to hold. The word, the file under the score, animate away.

Full time — FULL FRAME · TWO / TWO

5.4.2 Full time. The final score from the scoreboard's team logo dividers and divided by colours and four Player's name and **Heading** shows a full time card.



Score board — CLOSE UP · ON SCREEN

5.4.3 Score board. Scores in each of the quarters as the scoreboard's Stats. Extra quarters, scores showing time, club colours or the competition's.

Best of night — CLOSE UP · ON SCREEN

5.4.4 Best of night. Each court after the last game, where the night has been and the results, and the Sponsor sponsors in a slip. Options the media for sponsorship at off.

> **Cards sit in the space between the title and the scoreboard and stroke to fit when they have more in them than fits.**
>
> CHAPTER 5 · THE GRAPHICS

5.5 The competition

Ladder — CLOSE UP · ON SCREEN

5.5.1 Ladder. The division's ladder with the Pools places, rowed and a key under it. Options whether rows, logo on or off, light or dark more logo.

Results — CLOSE UP · ON SCREEN

5.5.2 Results. The scene round one to games are final, with scores. Options round option applies.

Finals — FULL FRAME · TWO / TWO

5.5.3 Finals. Who plays whom, week by week in the division's Pools format, with the scores and the winner highlighted. Options whether to show the bracket or the ladder, and the round, name shown, colours and order only the holding and the first under it.

Round fixtures — CLOSE UP · ON SCREEN

5.5.4 Round fixtures. The round's games across all divisions with **Round** and **Round** after next, you, it's a number or a name. Key, the round.

Top shooters — CLOSE UP · ON SCREEN

5.5.5 Top shooters. The season's leading shooters for a division. Options order by goals or average, how many shooters, a sponsor, fonts.

Competition logo — CLOSE UP · ON SCREEN

5.5.6 Competition logo animation. Sponsors competition.gov, competition logo, and the Studio's own sponsors in and out. Options and tweaks, about 5 seconds, only sponsors, and a duration time.

5.6 The breaks

Sponsor card — CLOSE UP · ON SCREEN

5.6.1 Sponsor card. One sponsor's logo, tagline, offer and a QR code to their website, taking turns by list. Options one sponsor always, which take rotate, animate away.

PIVT and FirstPass Studio — CLOSE UP · ON SCREEN

5.6.2 PIVT and FirstPass Studio ad. A 15-second one-ad set for a break, with nothing said. A Sponsor can take a slot in the line. Options **Show ad** to hold.

5.7 Checking the outputs

Test pattern — FULL FRAME · TWO / TWO

5.7.1 Test pattern. Colours, fine lines, ramps and soft gradients to check the monitor's picking and its key and to and weather the stream. Never for air.

CHAPTER 6 OF 11

Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

1 PlayHQ · 2 Names on the graphics · 3 Game rules · 4 Finals · 5 Game day · 6 Colours and font · 7 Logos · 8 Live viewers · 9 Templates, export and more

Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.

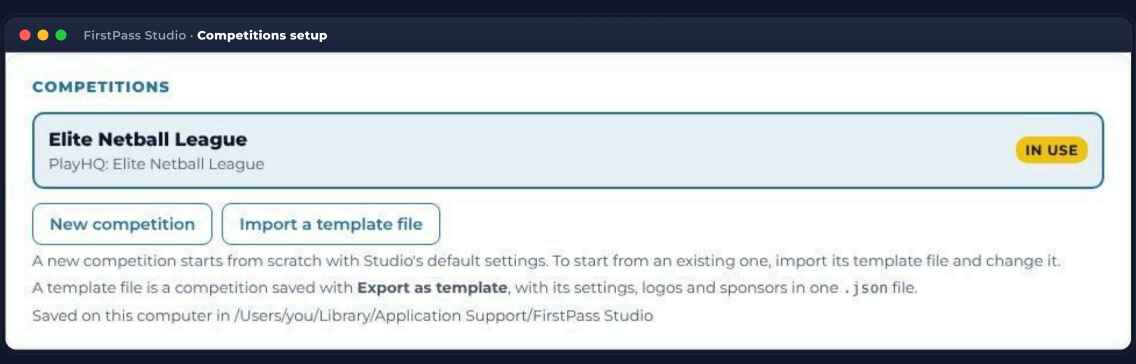


FIG 6.1 The competitions list. Start from a template file (the repository has Thunder League, AMMNA Nationals, GIG WANL and West Coast Thunder) or press New competition for Studio's defaults.

6.1 PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.

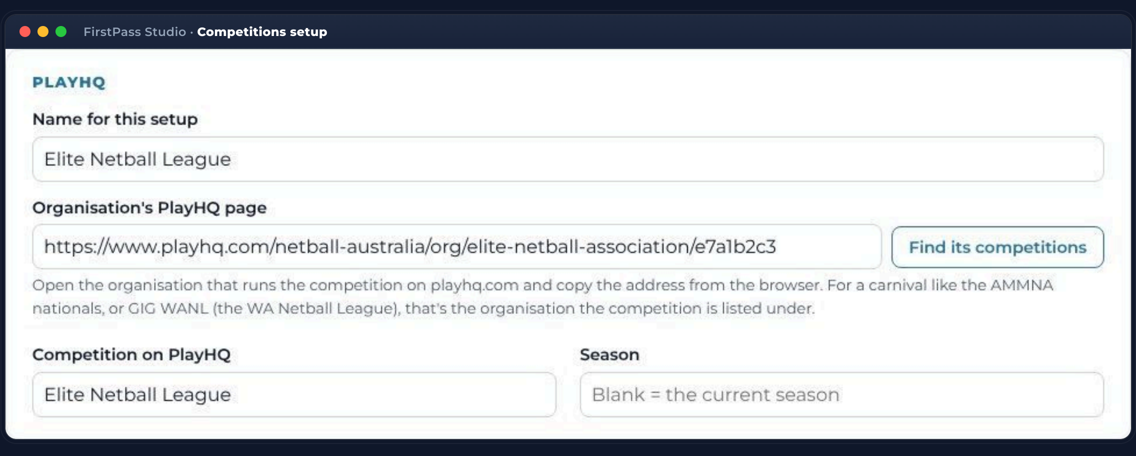


FIG 6.2 The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

6.2 Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.

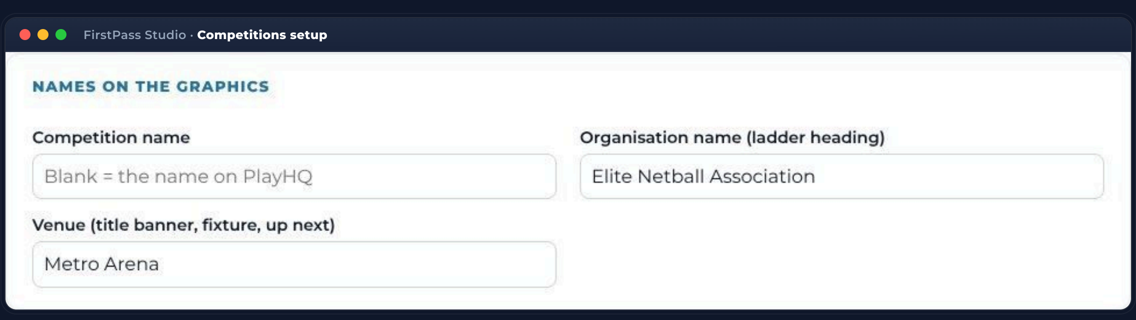


FIG 6.3 Names on the graphics.

6.3 Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.

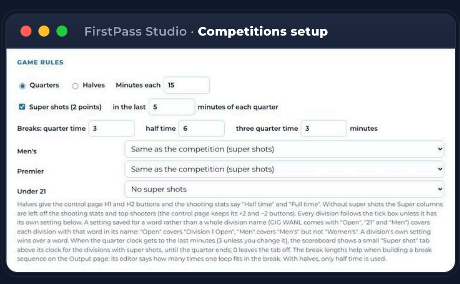


FIG 6.4 Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

6.4 Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control.

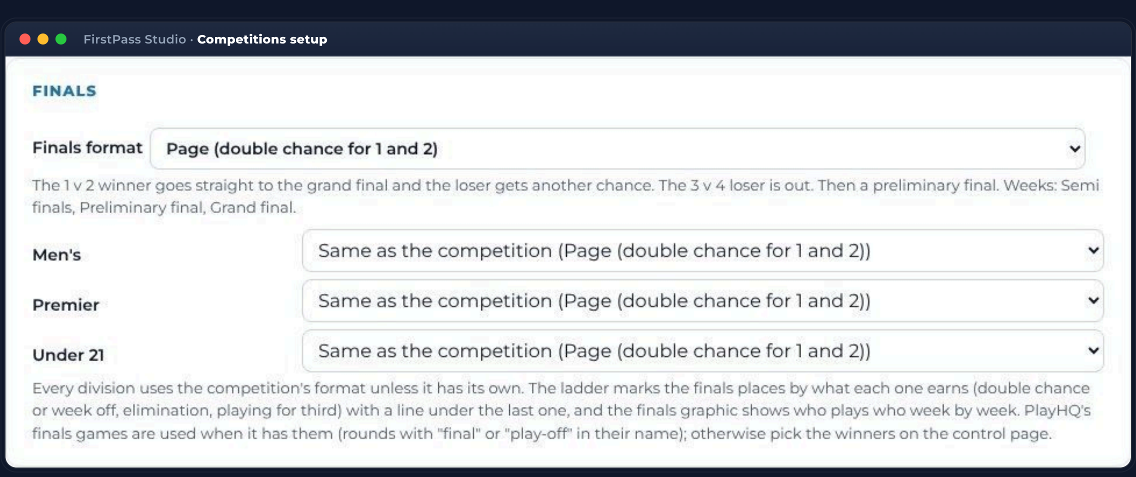


FIG 6.5 Finals formats, per division where they differ.

6.5 Game day

The court Studio follows, whether it picks the game automatically, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead "this round" is found. These are starting points: Scoreboard control can change the court, pick a game by hand or force a round on the night. On a carnival day with several rounds, this round moves on as each round's first game comes up.

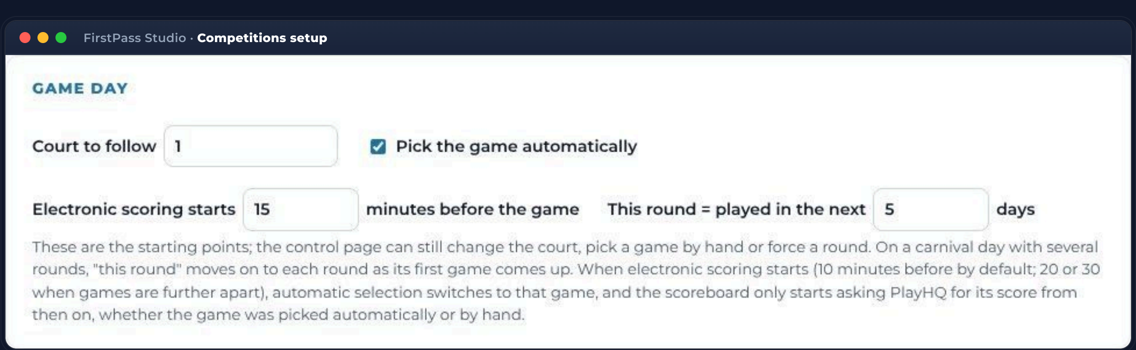


FIG 6.6 Game day.

6.6 Colours and font

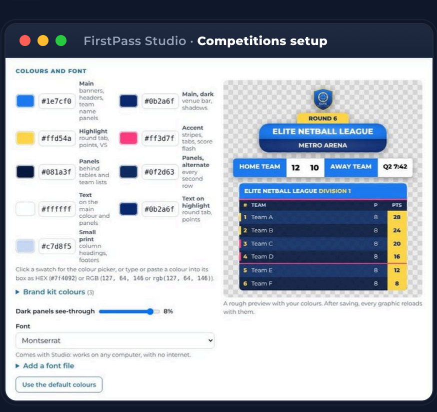


FIG 6.7 Colours and font, with the preview.

The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one per weight or a variable font) and it travels with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.

6.7 Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the branding logos, above the title with its option on, in the corner of the ladder and in the logo animations. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 21U" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash; without one the competition's colours are used.

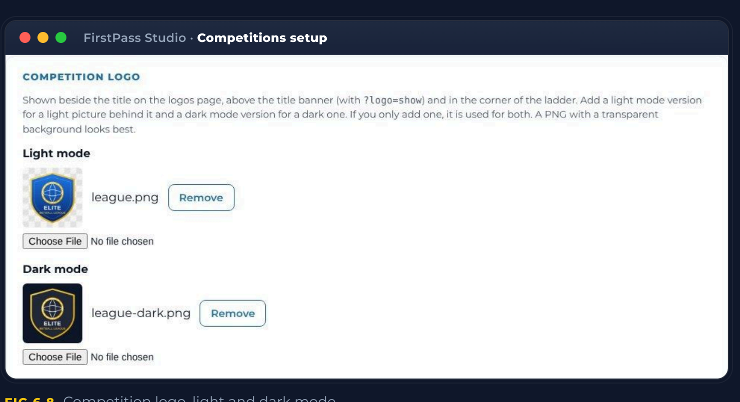


FIG 6.8 Competition logo, light and dark mode.

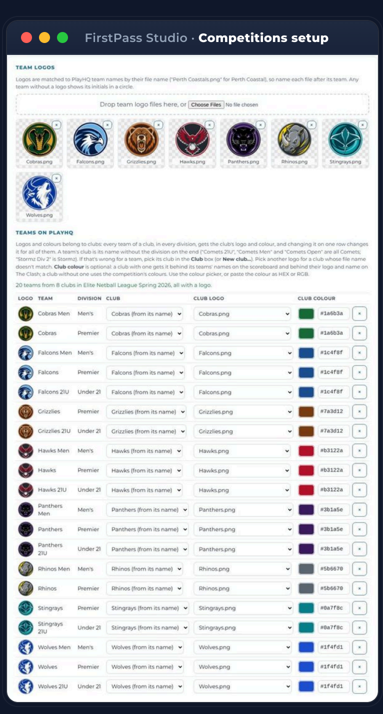


FIG 6.9 Team logos and the teams on PlayHQ, grouped by club with a club colour each.

> **Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour.**
>
> CHAPTER 6 · COMPETITIONS SETUP

6.8 Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.

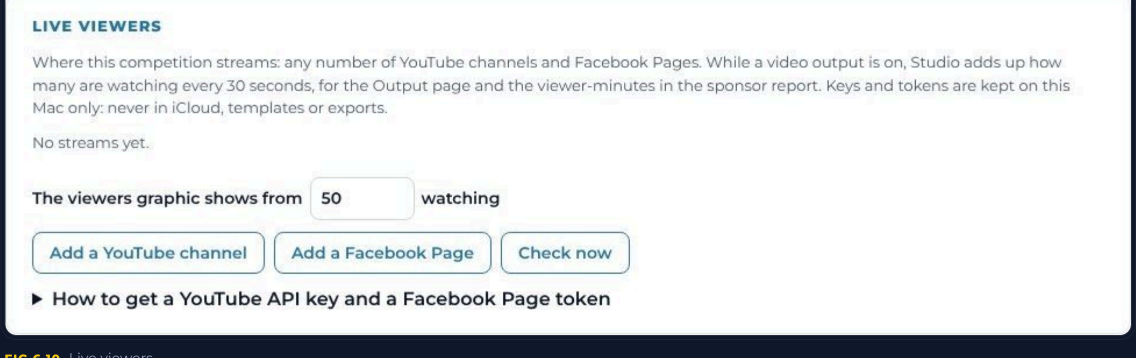


FIG 6.10 Live viewers.

6.9 Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.

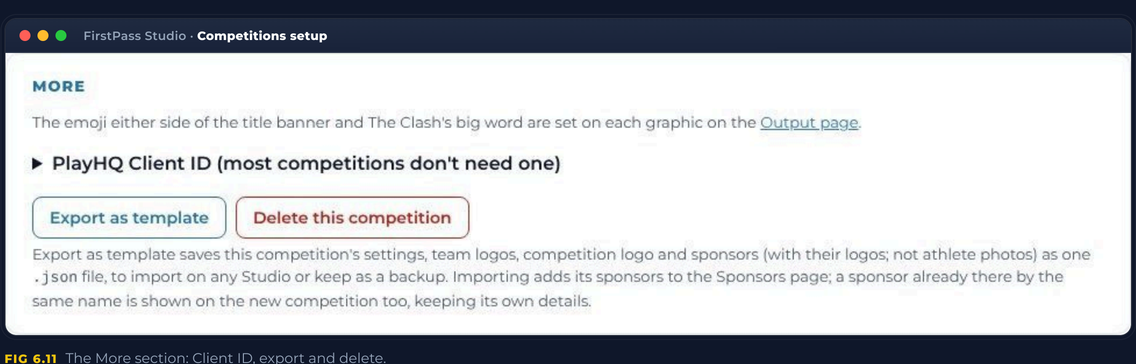


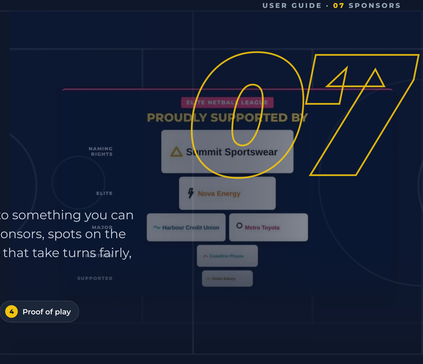
FIG 6.11 The More section: Client ID, export and delete.

CHAPTER 7 OF 11

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

1 Tiers · 2 A sponsor · 3 Spots and moments · 4 Proof of play



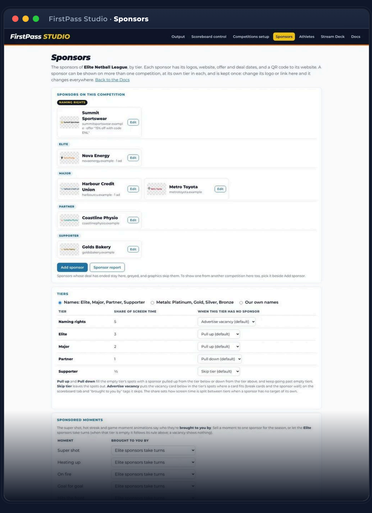


FIG 7.1

The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

7.1 Tiers

Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

> **The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.**
>
> CHAPTER 7 · SPONSORS

7.2 A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.

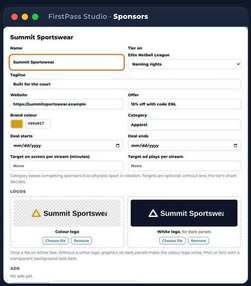


FIG 7.2 The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

7.3 Spots and moments

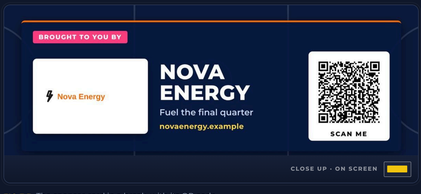


FIG 7.3 The sponsor card in a break, with its QR code.

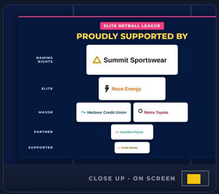


FIG 7.4 The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

7.4 Proof of play

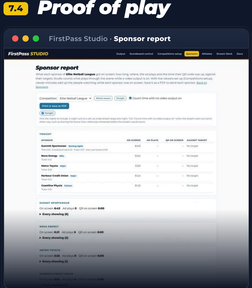


FIG 7.5 The sponsor report for the sample night.

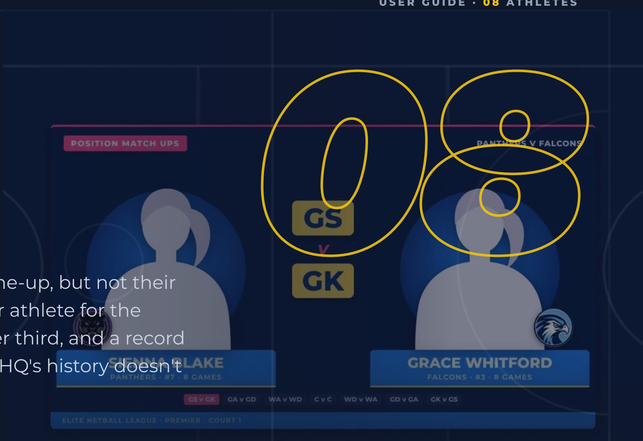
While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal.

CHAPTER 8 OF 11

Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

1. Photos
2. Positions and games played



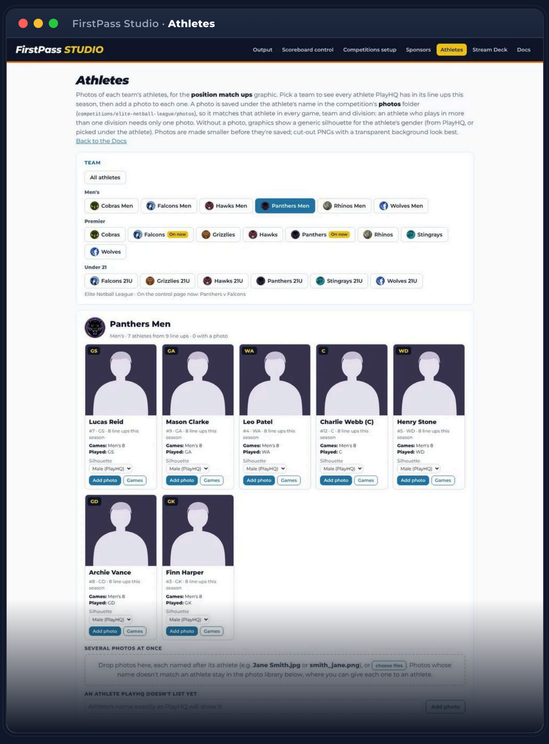


FIG 8.1

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

8.1 Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

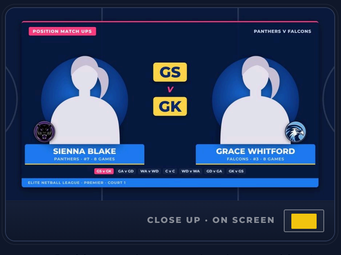


FIG 8.2 Position match ups use the photos, pairing by pairing.

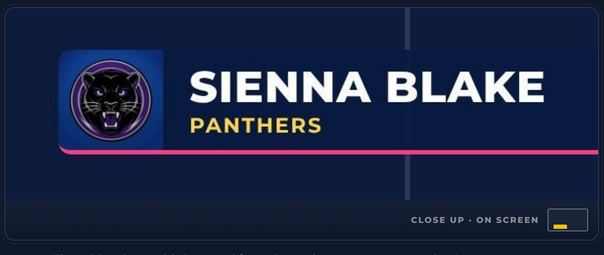


FIG 8.3 The athlete lower third, named from the register as you type on the Output page.

> ***A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo.***
>
> CHAPTER 8 · ATHLETES

8.2 Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.

FIG 8.4 An athlete's positions and games played.

CHAPTER 9 OF 11

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link.

1. The plugin
2. Links for everything else
3. Timed keys and the icon pack

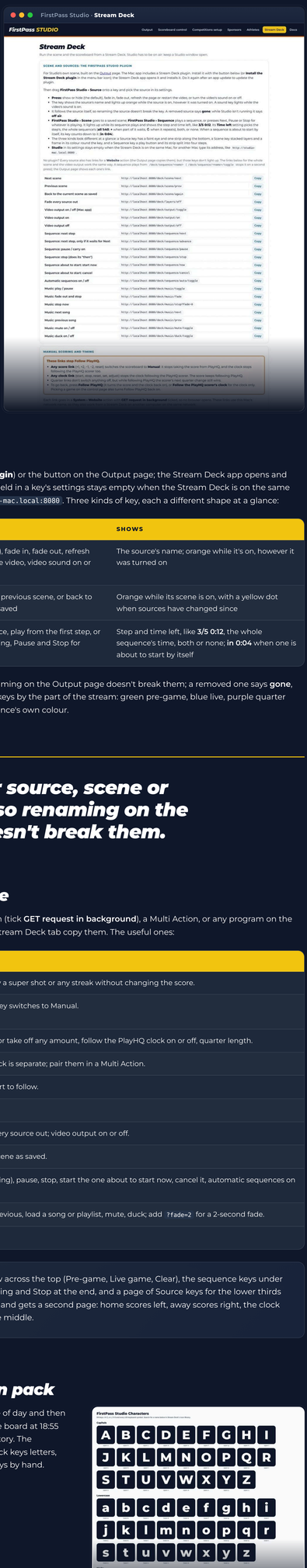


FIG 9.1

The Stream Deck tab: the plugin, the scoring and clock links, and the full reference below them.

9.1 The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Three kinds of key, each a different shape at a glance:

KEY	LOOKS	PRESS	SHOWS
Source	A faint eye and one strip along the bottom	Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off	The source's name; orange while it's on, however it was turned on
Scene	Stacked layers and a frame in its colour	Go to a scene, next or previous scene, or back to the current scene as saved	Orange while its scene is on, with a yellow dot when sources have changed since
Sequence	A play button and a strip in four steps	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**. Colour groups keys by the part of the stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads, or the scene's or sequence's own colour.

> ***Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them.***
>
> CHAPTER 9 · STREAM DECK

9.2 Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Stream Deck tab copy them. The useful ones:

GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.

A LAYOUT THAT WORKS Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

9.3 Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.

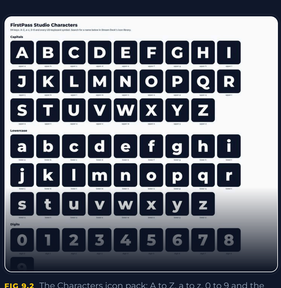


FIG 9.2 The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.

CHAPTER 10 OF 11

Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

1 The 1P menu
2 The data folder
3 Settings in iCloud Drive
4 Backup and restore
5 Updates
6 Links and other devices
7 When something isn't right
8 Data and bandwidth

10.1 The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up...** and **Restore...**; **Open the debug log**, **Restart Studio**, **Check for updates**; **Open at login**, **Let other devices change things**, **What's new...** and Quit.

10.2 The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

10.3 Settings in iCloud Drive

Settings folder > Move to iCloud Drive... moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac...** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac...** undoes the move. Studio goes off air for a moment for each and comes straight back.

10.4 Backup and restore

Back up... saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore...** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. Make a backup after setting up a season and before trying anything new on the Output page.

10.5 Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

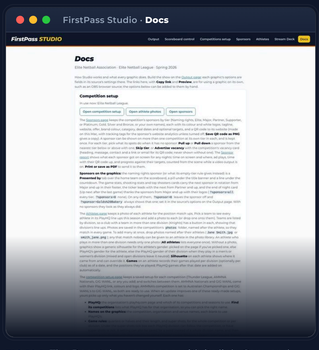


FIG 10.1 The Docs tab: every page with its link, Copy and Preview, the page options, and the notes this guide draws on.

> **Updates download by themselves and install when the app quits, never mid-stream.**
>
> CHAPTER 10 · SETTINGS AND CARE

10.6 Links and other devices

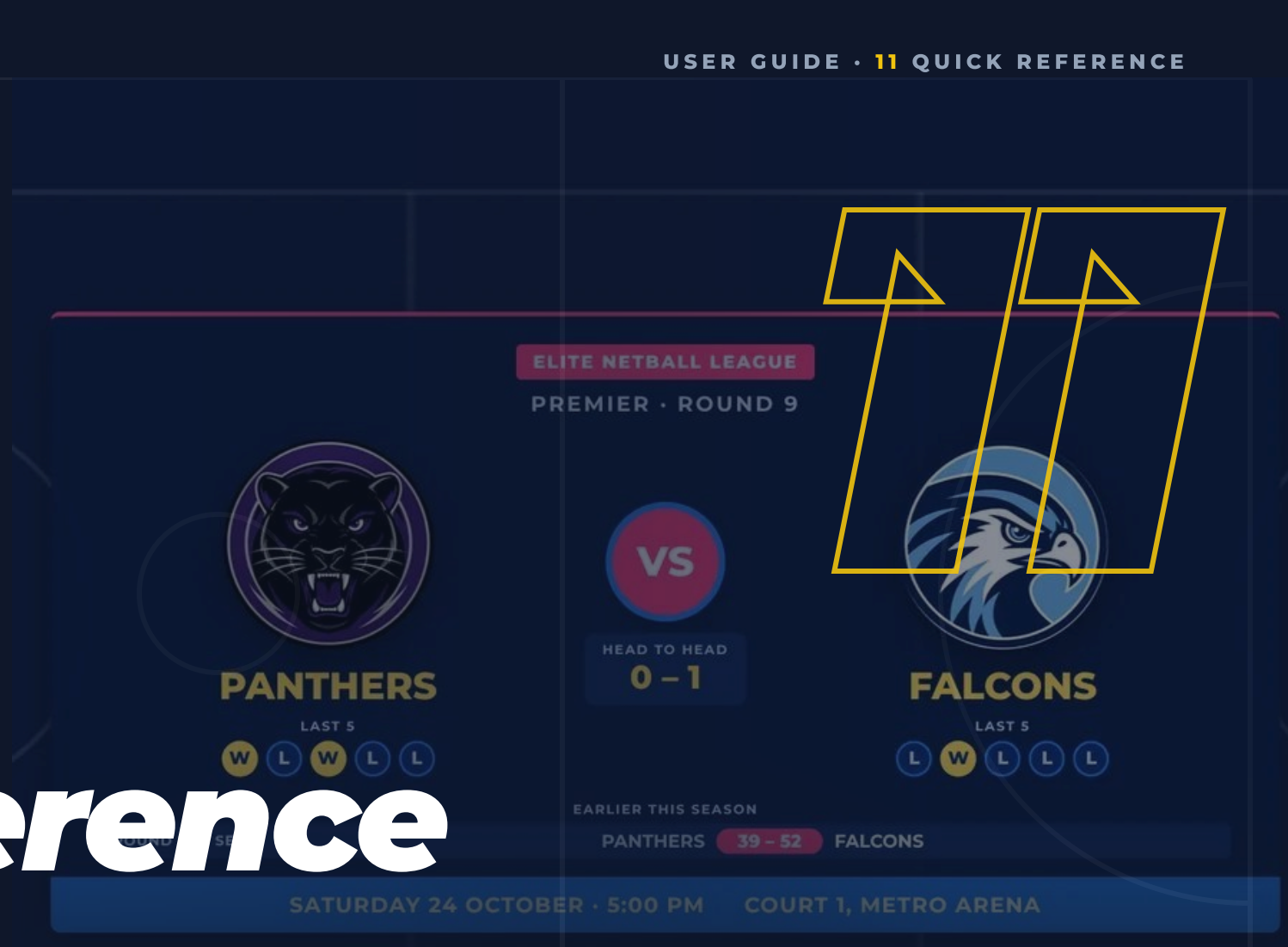
Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. **Let other devices change things** decides whether they can also change scores, the scene, sequences and settings; untick it on a network you don't control, or block incoming connections to FirstPass Studio in the Mac's firewall.

10.7 When something isn't right

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering. Switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong round or game is on	Force the round on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

10.8 Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game (or as set, down to one a second), the ticker's other courts every 30 seconds, the fixture about once a minute for automatic selection, and the rest once when a graphic loads. Finished games are read once and kept. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth.



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Quick reference

1 Keyboard
2 Stream night in nine lines
3 Where things are set

11.1 Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘7 go to Output, Scoreboard control, Competitions setup, Sponsors, Athletes, Stream Deck, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

11.2 Stream night in nine lines

01
Open Studio from the 1P menu and keep the Output page open all night.

02
Check the competition in use and the court on Competitions setup.

03
On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.

04
Turn on the video output your switcher takes and show one graphic to see it land.

05
Play a few seconds of music and an animation with sound.

06
Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.

07
Go to the Live game scene and let the sequences run the breaks.

08
Pick finals winners as they happen; turn on lower thirds when people appear.

09
Press Stream close, turn the output off, and send the sponsor report.

11.3 Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Keys and links	Stream Deck
Backups, iCloud, updates, the debug log	The 1P menu

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

FirstPass STUDIO

MADE BY PIVT

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